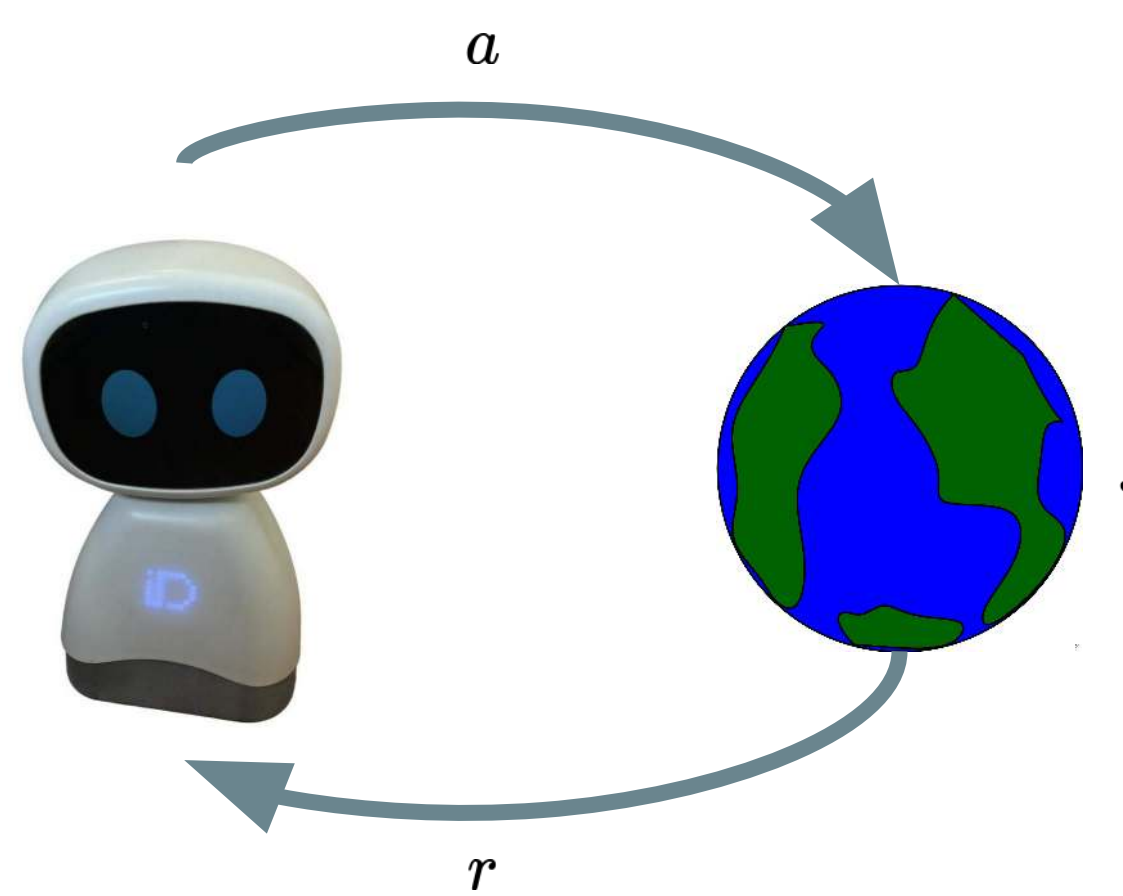


Does the Q-learning algorithm solve feudal hierarchies?

Reinforcement Learning

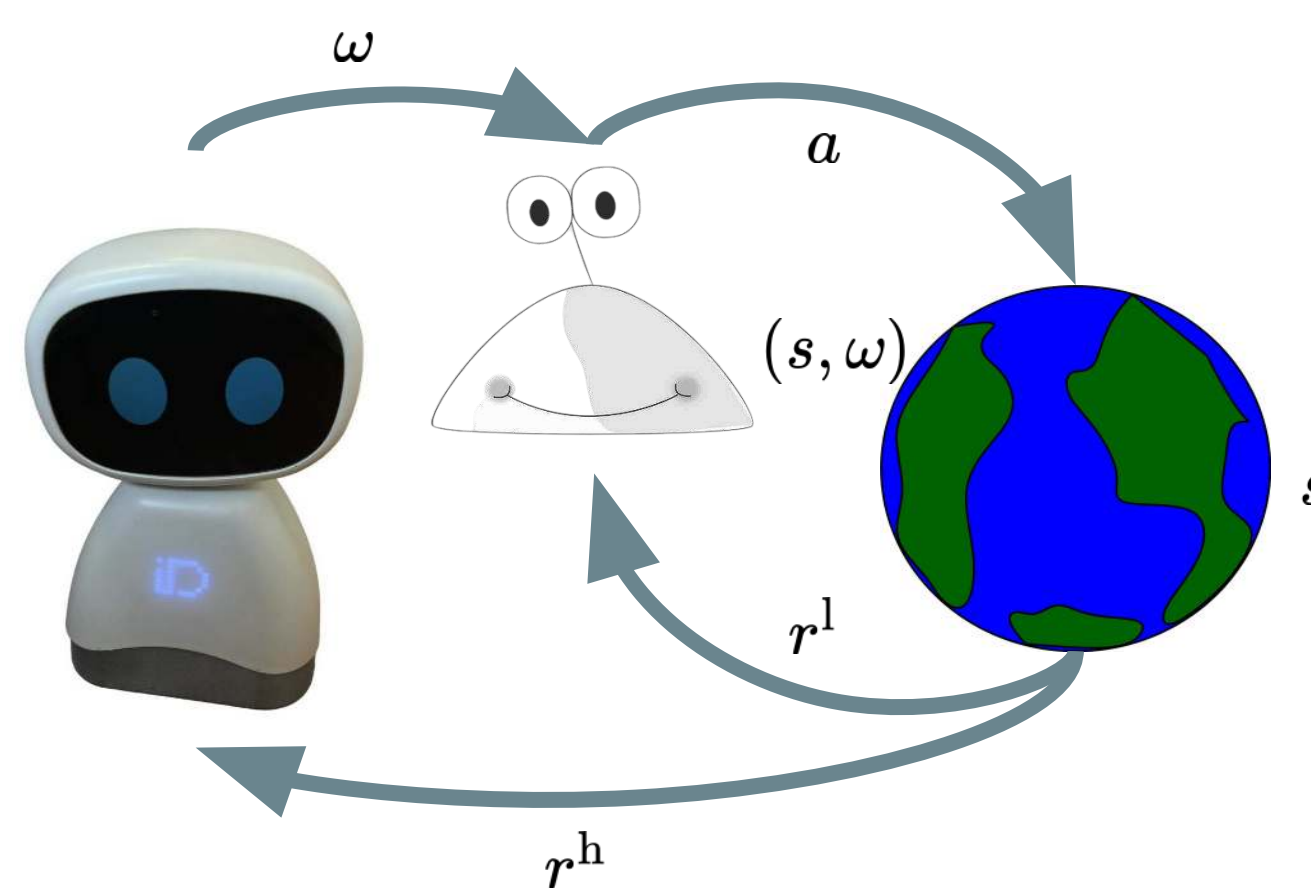
MDP.	$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$
Policy.	$\pi : \mathcal{S} \rightarrow \mathcal{A}$
Q-function.	$q^\pi = \mathbb{E} [\sum_{t=0}^{\infty} \gamma^t r_t]$
Objective.	$q^* = \max_{\pi} q^\pi$
Q-learning.	$q_{t+1} = (1 - \alpha)q_t + \alpha (r_t + \gamma \max_{q'} q'_t)$



Remark 1. The Q-learning algorithm converges to the optimal policy in stationary MDPs.

Feudal Hierarchies

High-level MDP.	$\mathcal{M}^h = (\mathcal{S}, \Omega, \mathcal{P}^h(\pi^l), \mathcal{R}^h(\pi^l))$
Low-level MDP.	$\mathcal{M}^l = (\mathcal{S} \times \Omega, \mathcal{A}, \mathcal{P}^l, \mathcal{R}^l)$



Remark 2. The high-level MDP depends on the low-level policy. The MDP is non-stationary.

Theorem 1.

Despite non-stationary, the high-level MDP converges if the low-level policy converges.

Proof.

Each component of the MDP is a continuous function of the policy. Equivalently, we conclude the MDP converges as the policy converges.

Theorem 2.

The Q-learning algorithm converges in non-stationary convergent MDPs.

Proof.

The errors are $\Delta_{t+1} = (1 - \alpha)\Delta_t + \alpha F_t$, where $\Delta_t = q_t - q^*$ and $F_t = r_t + \gamma \max_{q'} q'_t - q^*$. We show that $\|F_t\| \leq \gamma\|\Delta_t\| + \epsilon_t$, where $\epsilon_t \rightarrow 0$. Therefore, $\Delta_t \rightarrow 0$ and $q_t \rightarrow q^*$.

Conclusion. Q-learning converges to the optimal solution in feudal hierarchies.

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